Optimized production of ultracold ground-state molecules: Stabilization employing potentials with ion-pair character and strong spin-orbit coupling

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We discuss the production of ultracold molecules in their electronic ground state by photoassociation employing electronically excited states with ion-pair character and strong spin-orbit interaction. [illegible]

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I. INTRODUCTION

Photoassociation, which is the forming of molecules from ultracold atoms using laser light [1], is a prime candidate for coherent control which utilizes the wave nature of matter in order to steer a process, such as the formation of a chemical bond, toward a desired target [2,3]. At very low temperature, the delicate balance of constructive and destructive interference between different quantum pathways is not hampered by thermal averaging. The basic tool for coherent control are short laser pulses that can be shaped in their amplitude, phase, and polarization. They can drive both adiabatic and nonadiabatic photoassociation dynamics [4].

A particular feature of photoassociation at low temperatures is the existence of an atom pair at very large interatomic separations [1]. [illegible]

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